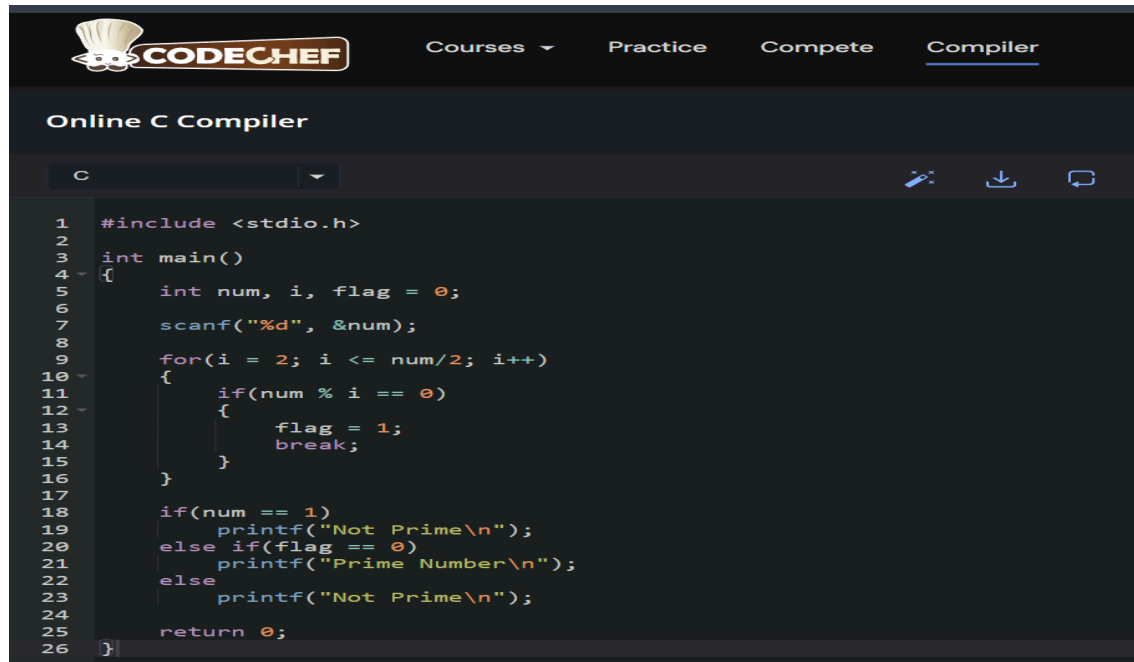


COURSE	: COMPUTER PROGRAMMING (CS F111)
COMPONENT	: Practical Week 4: Looping statements

1. Write program to check if a number is prime or not using for loop.

CODE:

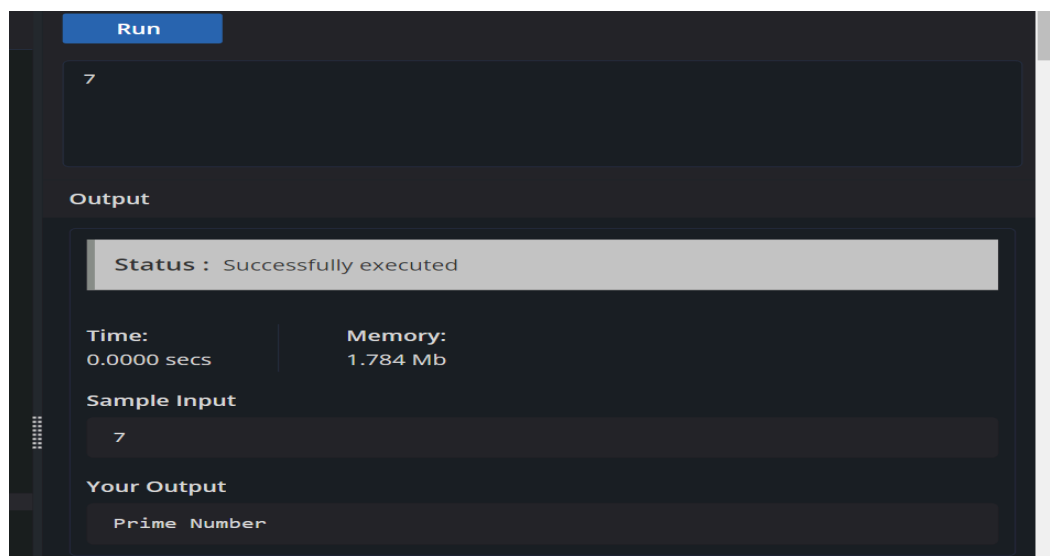


```

1  #include <stdio.h>
2
3  int main()
4  {
5      int num, i, flag = 0;
6      scanf("%d", &num);
7
8      for(i = 2; i <= num/2; i++)
9      {
10         if(num % i == 0)
11         {
12             flag = 1;
13             break;
14         }
15     }
16
17     if(num == 1)
18         printf("Not Prime\n");
19     else if(flag == 0)
20         printf("Prime Number\n");
21     else
22         printf("Not Prime\n");
23
24     return 0;
25 }

```

OUTPUT:



Run

7

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.784 Mb

Sample Input

7

Your Output

Prime Number

2. Write a program to find if a number is a magic number using do while loop

Hint: Find the sum of the digits of the given number. If the product of the sum and the reverse number of the sum is the given number then the number is the magic number.

ex: 1729: Sum of digits: $1+7+2+9=19$

the reverse of sum: 91

product of sum and reverse of sum: $19*91=1729$

hence it is magic number.

CODE:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int num, temp, sum = 0, digit;
6      int rev = 0, product;
7
8      scanf("%d", &num);
9
10     temp = num;
11
12     // Find sum of digits using do-while
13     do
14     {
15         digit = temp % 10;
16         sum = sum + digit;
17         temp = temp / 10;
18     } while(temp != 0);
19
20     // Find reverse of the sum
21     temp = sum;
22
23     do
24     {
25         digit = temp % 10;
26         rev = rev * 10 + digit;
27         temp = temp / 10;
28     } while(temp != 0);
29
30     product = sum * rev;
31
32     if(product == num)
33         printf("Magic Number\n");
34     else
35         printf("Not Magic Number\n");
36
37     return 0;
38 }
```

OUTPUT:

The screenshot shows a code execution environment. At the top, a 'Run' button is visible. Below it, the input '1729' is entered. The 'Output' section shows 'Status : Successfully executed'. Below that, a table displays 'Time: 0.0000 secs' and 'Memory: 1.696 Mb'. Under 'Sample Input', '1729' is shown. Finally, under 'Your Output', 'Magic Number' is displayed.

Time:	Memory:
0.0000 secs	1.696 Mb

Sample Input

1729

Your Output

Magic Number

3. Write a program to print factorial of all the numbers from 1 to n **using nested loop**.

Eg. If user input is 5

Then the program prints

Factorial of 1= 1

Factorial of 2= 2

Factorial of 3= 6

Factorial of 4=24

Factorial of 5=120.

CODE:

```
C
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j;
6      long long factorial;
7
8      scanf("%d", &n);
9
10     for(i = 1; i <= n; i++)
11     {
12         factorial = 1;
13
14         for(j = 1; j <= i; j++)
15         {
16             factorial = factorial * j;
17         }
18         printf("Factorial of %d = %lld\n", i, factorial);
19     }
20
21     return 0;
22 }
23 }
```

OUTPUT:

```
Run
5

Output
Status : Successfully executed

Time: 0.0000 secs    Memory: 1.656 Mb

Sample Input
5

Your Output
Factorial of 1 = 1
Factorial of 2 = 2
Factorial of 3 = 6
Factorial of 4 = 24
Factorial of 5 = 120
```

4. Write a program to print Fibonacci series using loop.

0 1 1 2 3 5 8 13 21

Int a=0, b=1,c;

c=a+b;

a=b;

b=c;

CODE:

```
C
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i;
6      int a = 0, b = 1, c;
7
8      scanf("%d", &n);
9
10     printf("%d %d ", a, b);
11
12     for(i = 3; i <= n; i++)
13     {
14         c = a + b;
15         printf("%d ", c);
16
17         a = b;
18         b = c;
19     }
20
21     return 0;
22 }
```

OUTPUT:

